The image of the big bang as a cosmic explosion ejecting the material contents of the universe like shrapnel from an exploding bomb is a useful one to bear in mind, but it is a little misleading. When a bomb explodes, it does so at a particular location in space and at a particular moment in time. Its contents are ejected into the surrounding space. In the big bang, there is no surrounding space. As we devolve the universe backward toward the beginning, the squeezing together of all material content occurs because all of space is shrinking. The orange-size, the pea-size, the grain of sand-size devolution describes the whole of the universe—not something within the universe. Carrying on to the beginning, there is simply no space outside the primordial pinpoint grenade. Instead, the big bang is the eruption of compressed space whose unfurling, like a tidal wave, carries along matter and energy even to this day.

Is General Relativity Right?

No deviations from the predictions of general relativity have been found in experiments performed with our present level of technology. Only time will tell if greater experimental precision will ultimately uncover some, thereby showing this theory, too, to be only an approximate description of how nature actually works. The systematic testing of theories to greater and greater levels of accuracy is, certainly, one of the ways science progresses, but it is not the only way. In fact, we have already seen this: The search for a new theory of gravity was initiated, not by an experimental refutation of Newton's theory, but rather by the conflict of Newtonian gravity with another theory—special relativity. It was only after the discovery of general relativity as a competing theory of gravity that experimental flaws in Newton's theory were identified by seeking out tiny but measurable ways in which the two theories differ. Thus, internal theoretical inconsistencies can play as pivotal a role in driving progress as do experimental data.

For the last half century, physics has been faced with still another theoretical conflict whose severity is on par with that between special relativity and Newtonian gravity. General relativity appears to be fundamentally incompatible with another extremely well-tested theory: quantum mechanics. Regarding the material covered in this chapter, the conflict prevents physicists from understanding what really happens to space, time, and matter when crushed together fully at the